

ABHIJIT KUMAR

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Summary

Computer Science & Artificial Intelligence undergraduate with hands-on experience building and deploying AI-powered applications across NLP, computer vision, machine learning, and full-stack development. Developed production-grade AI systems using LLMs, RAG pipelines, YOLOv8, and XGBoost; integrated AI models into real-world applications with clean, maintainable code. Co-authored multi-institutional AI research manuscript; national hackathon winner (1st of 150+ teams); Top-10 AI Finalist at IIT Kanpur Techkriti 2025. Comfortable working independently in fast-paced, remote-first environments.

Skills

Languages: Python, Java, C#

AI & Machine Learning: Scikit-learn, XGBoost, YOLOv8, OpenCV, Pandas, NumPy, RAG Pipelines, LLM Integration, Prompt Engineering, Sentence-BERT, Embeddings

NLP & Computer Vision: Document Summarization, Text Classification, Semantic Search, OCR (Tesseract), Object Detection, Real-Time Video Analysis

Backend & APIs: FastAPI, Django, Node.js, RESTful APIs, WebSockets, JWT Authentication, AES-128 Encryption

Frontend: React.js, HTML5, CSS3, Component-Based UI, Responsive Web Design, WebSocket Dashboards

Databases & Search: PostgreSQL, MySQL, MongoDB, SQLite, FAISS, SQL Server, SQLite FTS5

Tools & Practices: Git, AWS, Linux, Agile, Debugging, Unit Testing, Technical Documentation, VS Code, Jupyter Notebook, Google Colab

Projects & Research

Breath.exe — AI-Powered Wellness Classification System | Multi-Institutional Research 2024 – Present

Parul University (India) · ITMO University (Russia) · University of Wrocław (Poland) | Manuscript in Preparation

- Developed, trained, and deployed a multi-domain AI classification system across 80,149 samples and 28 engineered features — benchmarked five algorithms with XGBoost achieving top performance across Accuracy, F1-Macro, and ROC-AUC — and built a production-ready full-stack application (FastAPI + React.js) with local-first privacy, contributing to a co-authored manuscript with ITMO University (Russia) and University of Wrocław (Poland).

A.R.Y.A — AI-Powered NLP Document Automation | Cisco Challenge Track, 1st Place (150+ Teams) Oct 2025

Stack: Python, NLP Pipeline, Document Summarization, FastAPI, React.js, Git

- Led a 3-member team to design and deploy an AI-powered NLP system for automated document categorization and summarization within a 24-hour Agile sprint — reduced manual review effort by ~60% through intelligent text classification and extraction, ranking 1st nationally among 150+ teams in the Cisco Challenge Track.

RepoIR — AI Semantic Search & Document Intelligence Platform Aug – Dec 2025

Stack: Python, FAISS, Sentence-BERT, SQLite FTS5, FastAPI, React.js, Tesseract OCR, JWT, AES-128

- Built and deployed a production-grade AI semantic search system integrating FAISS vector search (Sentence-BERT dense embeddings), SQLite FTS5 full-text indexing, and an OCR ingestion pipeline (Tesseract) — cutting document processing effort by ~70% and achieving ~3x retrieval throughput improvement — with 8+ secured REST APIs (JWT + AES-128) enabling sub-200ms semantic search across documents.

UrbanFlowAI — Real-Time AI Vision & Analytics Platform May 2026

Stack: Python, YOLOv8, OpenCV, FastAPI, React.js, PostgreSQL, WebSockets

- Deployed an end-to-end real-time AI traffic monitoring system — deploying YOLOv8 with OpenCV for multi-vehicle detection and congestion classification, optimizing the live video inference pipeline, and integrating emergency route prioritization — backed by a FastAPI backend processing real-time model outputs, PostgreSQL analytics storage, and a React.js WebSocket dashboard delivering live data streaming across urban junctions.

Experience

C# .NET Developer Trainee — Cognizant Apr 2026 | Vadodara, Gujarat

- Developed backend modules, RESTful service layers, and relational database solutions using C#, .NET Framework, and SQL Server; applied OOP design patterns, unit testing, and structured debugging within enterprise Agile SDLC workflows.
- Collaborated across sprint planning, daily standups, retrospectives, and peer code reviews using Git-based version control; delivered technical documentation for all implemented backend modules.

Education

B.Tech — Computer Science & Engineering (Artificial Intelligence) Parul University, Vadodara, Gujarat 2023 – 2027

Achievements & Certifications

- 1st Place** — Cisco Challenge Track: Ranked 1st among 150+ national teams for an AI NLP automation product delivered in a 24-hour sprint.
- Top-10 Finalist** — AI Finance Hackathon, IIT Kanpur Techkriti 2025: Built an ML-based predictive analytics system for financial market forecasting.
- Silver Medalist** — Summer School Research Paper Award: Recognised for technical contribution to a multi-institutional AI research publication.
- Problem Solving:** 400+ LeetCode problems; 5-star Python & Java on **HackerRank**; 2-star **CodeChef**.
- Certifications:** AWS Cloud Foundations & Practitioner Essentials | IBM AI Fundamentals | GenAI 101 (nasscom) | Hashgraph Developer Course | Software Engineering Intern (HackerRank) | Green Skills & AI (Edunet)